Proof of Invariance for the Spinor Projection Operator Π

Theorem. The projection operator Π, defined via a rank-6 Levi-Civita contraction on a 12-vertex configuration in $\mathbb{R}^6$, is well-defined, orientation-consistent, and invariant under global spinor rotation $e^{i\theta}$.

Proof. Let the 12 vertices be embedded in $\mathbb{R}^6$. Complexify each coordinate pair to define spinors

$$z_{ij} = \hat{x}_{ij} + i\hat{y}_{ij} \in \mathbb{C}.$$

The global spinor rotation acts by

$$\Psi_\theta : z_{ij} \mapsto e^{i\theta} z_{ij}, \tag{1}$$

which induces a linear orthogonal transformation on each $\mathbb{R}^2$ coordinate plane.

The projection operator Π is constructed using the rank-6 Levi-Civita tensor ε_{ijklmn}. Under any transformation $O \in SO(6)$, the tensor transforms as

$$\varepsilon'_{abcdef} = \varepsilon_{ijklmn}\, O_{ia} O_{jb} O_{kc} O_{ld} O_{me} O_{nf} = \det(O)\, \varepsilon_{abcdef}. \tag{2}$$

Since the spinor rotation $e^{i\theta}$ corresponds to a block-diagonal product of $SO(2)$ rotations, the induced transformation $O_{\text{total}} \in SO(6)$ satisfies

$$\det(O_{\text{total}}) = 1.$$

Thus the Levi-Civita contraction is invariant, and therefore

$$\Pi \circ \Psi_\theta = \Pi.$$

The reduction from $\mathbb{R}^6$ to $\mathbb{R}^3$ follows the structure of the covering map

$$SU(2) \longrightarrow SO(3).$$

The spinor phase θ acts as the fiber of a Hopf-type fibration, restricting the effective degrees of freedom to a 3-dimensional subspace. Because the rotation is continuous and homotopic to the identity, orientation is preserved.

Hence, Π is well-defined, orientation-consistent, and independent of the choice of 6D embedding. $\square$